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Surface Color Properties of the Hayabusa2# Target Asteroid "Torifune" from Simultaneous Three-Filter Photometric Observations with the Seimei Telescope

Kazuya DOI[1], Seiko TAKAGI[2], Tomohiko SEKIGUCHI[3], Daisuke KURODA[4], Seitaro URAKAWA[5], Jin BENIYAMA[6,7], Masateru ISHIGURO[8], Jooyeon GEEM[9], Hiroki SENSYU[10]
[1]Sapporo City Higashitsukisamu Junior High School; [2]Hokkaido University; [3]Hokkaido University of Education; [4]Japan Spaceguard Association; [5]National Institute of Technology, Matsue College; [6]The University of Tokyo; [7]Observatoire de la Côte d'Azur; [8]Seoul National University; [9]Luleå University of Technology; [10]Chiba Institute of Technology

1. Introduction

【JAXA Hayabusa2# (SHARP)】

Small **H**azardous **A**steroid **R**econnaissance **P**robe
Extended mission after the sample-return from Ryugu (e.g. Tsuda+ 2025, Hirabayashi+ 2026)

Objectives:
1. Evolution of long-term deep space cruise technology
2. Exploration of fast rotator asteroid (1998 KY26)
3. Acquisition of science and technology for planetary defense

Targets:
1. Torifune (2001 CC21), **Flyby in July 5, 2026 (This year!)**
2. 1998 KY26, Rendezvous in July 2031

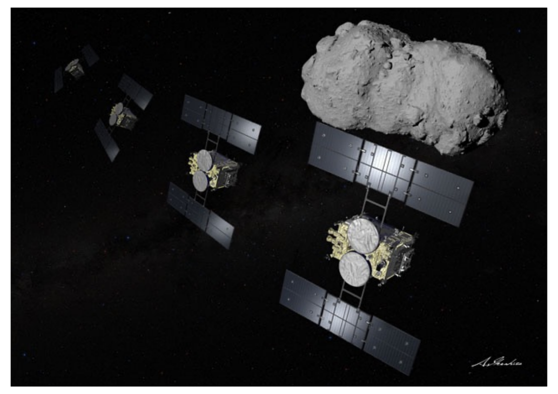


Fig. 1 Torifune flyby image of Hayabusa2 (JAXA).

【Super-close High-speed Flyby】

- Closest approach distance: 10-1 km (TBD by JAXA)
- Relative velocity: 5.25 km/s

→ Designed for the Ryugu rendezvous, can't focus, must close approach
→ Exposure time & gain must be **preset**
→ Only **one side** of Torifune can be observed during the flyby

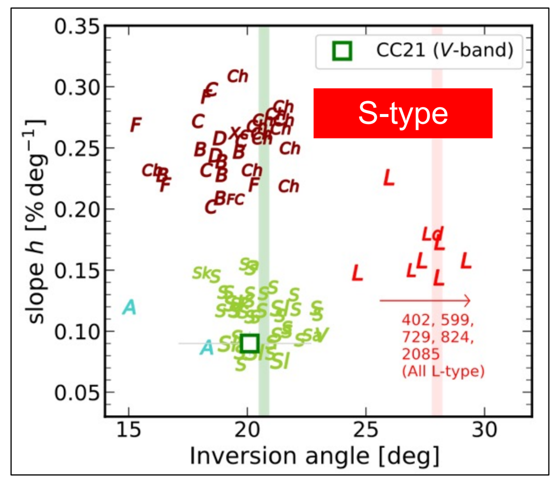


Fig. 2 Polarimetric properties of Torifune (CC21), consistent with S-type asteroids (Geem+ 2023).

【Previous observations】

- Polarimetric observation (Geem+ 2023)

→ **S-type** polarimetric properties, not L-type (Fig. 2)

- Spectroscopic observation (Geem+ 2023, Popescu+ 2025)

→ **Sq- (S-)type** reflectance spectrum (Fig. 3)
Sq-type: moderately space-weathered S-type

- Photometric observation (Popescu+ 2025)

→ **S-type** color-index; g-r, r-i, r-z_s (Fig. 4)
→ Small variation, large scale **surface color homogeneity**

- Light-curve observation (e.g. Fatka+ 2025, Popescu+ 2025)

→ **5.02 h** rotation period (Fig. 5)

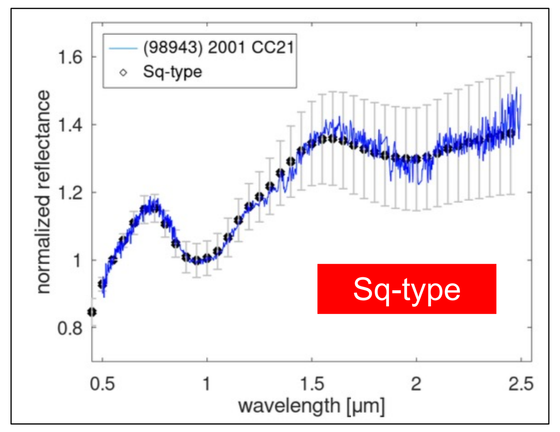


Fig. 3 Visible-NIR reflectance spectrum of Torifune ((98943) 2001 CC21), consistent with Sq- (S-)type asteroids (Popescu+ 2025).

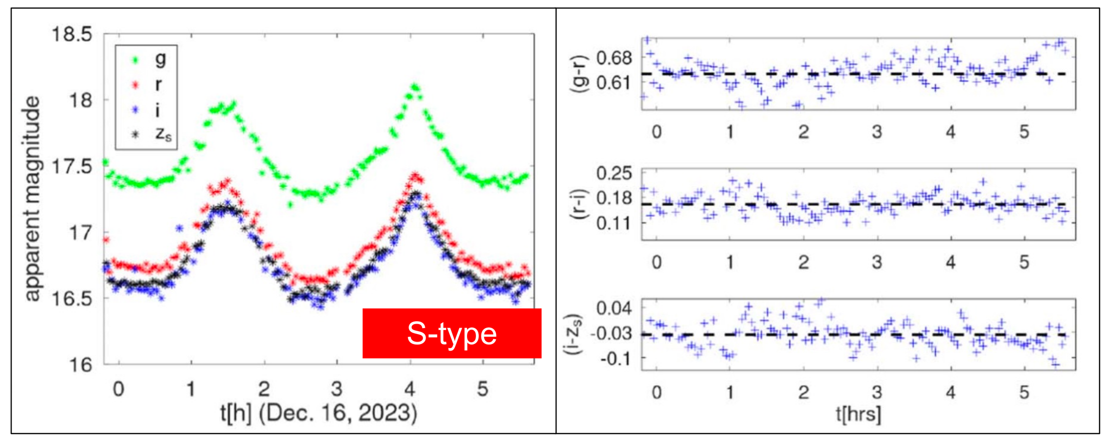


Fig. 4 left) Multi-color light-curve of Torifune. right) SDSS system filter color-index of Torifune, consistent with the values of S-type asteroids and showing small variation associated with its rotation (Popescu+ 2025).

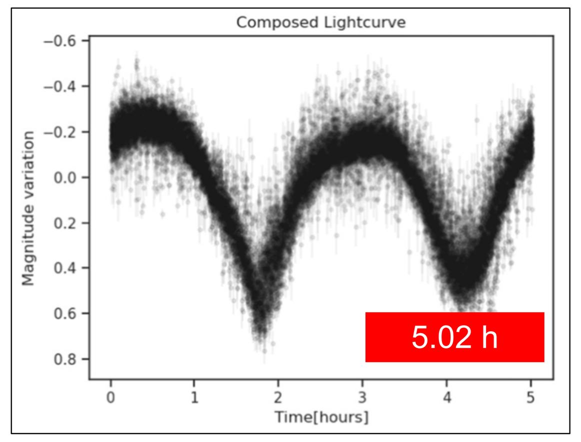


Fig. 5 Light-curve of Torifune (Popescu+ 2025).

【Significance of observations before spacecraft missions】

- **Technical**: Pre-setting the exposure time and gain of onboard cameras

→ Can't be properly determined unless the surface color **in advance**

- **Scientific**: Surface color properties of Torifune

→ **Surface color homogeneity or heterogeneity**
→ **Highest scientific side** for observation
→ Whether the one side information **represents Torifune's entire surface** (including un-observed side)

【Objectives】

Simultaneous three-filter photometry of Torifune before the Hayabusa2# flyby to derive its **surface color properties** with **high time resolution and high photometric accuracy**

2. Observations

【3.8 m Seimei telescope/TriCCS @Okayama】

- 2 proposals were approved by NAOJ
- The largest optical telescope in East Asia, optimal for observing faint asteroids (Fig. 6 left)
- Simultaneous three-filter photometry in Pan-STARRS system *g*, *r*, *i* or *z*-filter (Fig. 6 right, 7)

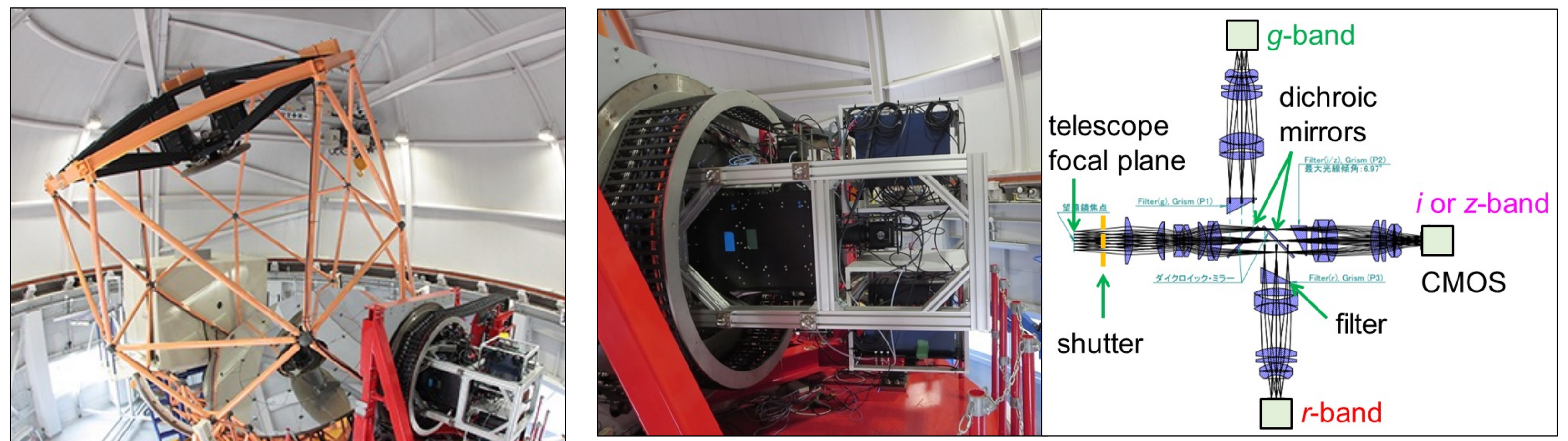


Fig. 6 left) Seimei telescope at Okayama observatory owned by Kyoto University. right) Tricolor CMOS Camera and Spectrograph (TriCCS), TriCCS optics in the imaging mode.

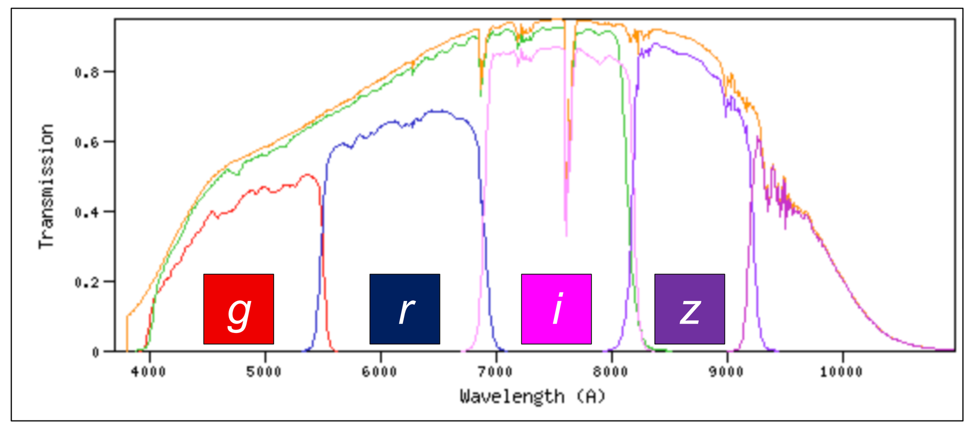


Fig. 7 Transmittance of Pan-STARRS system filter.

Tab. Observational list.

Date	Phase angle	Filter	Notes
Nov. 27, 2024	6°	*g, r, i*	Cloudy
Nov. 28, 2024	7°	*g, r, i*	Partly Cloudy
Oct. 22, 2025	20°	*g, r, i* & *g, r, z*	Partly Cloudy
Nov. 6, 2025	25°	*g, r, i*	Clear DDT

3. Results & Discussion

【Multi-color light-curve】

- **High time resolution** (exp. time 12 s) & **high photometric accuracy** (S/N > 100) data (Fig. 8)
- Consistent with the observational geometry model (Fig. 8, 9)

→ Brightest at the max. cross section (Fig. 9 left) and faintest at the min. cross section (Fig. 9 right)

- Fold 3 nights data with rotation period of 5.02 h (Fig. 10)

→ The rotational magnitude variation is consistent with the values calculated by the Hapke model

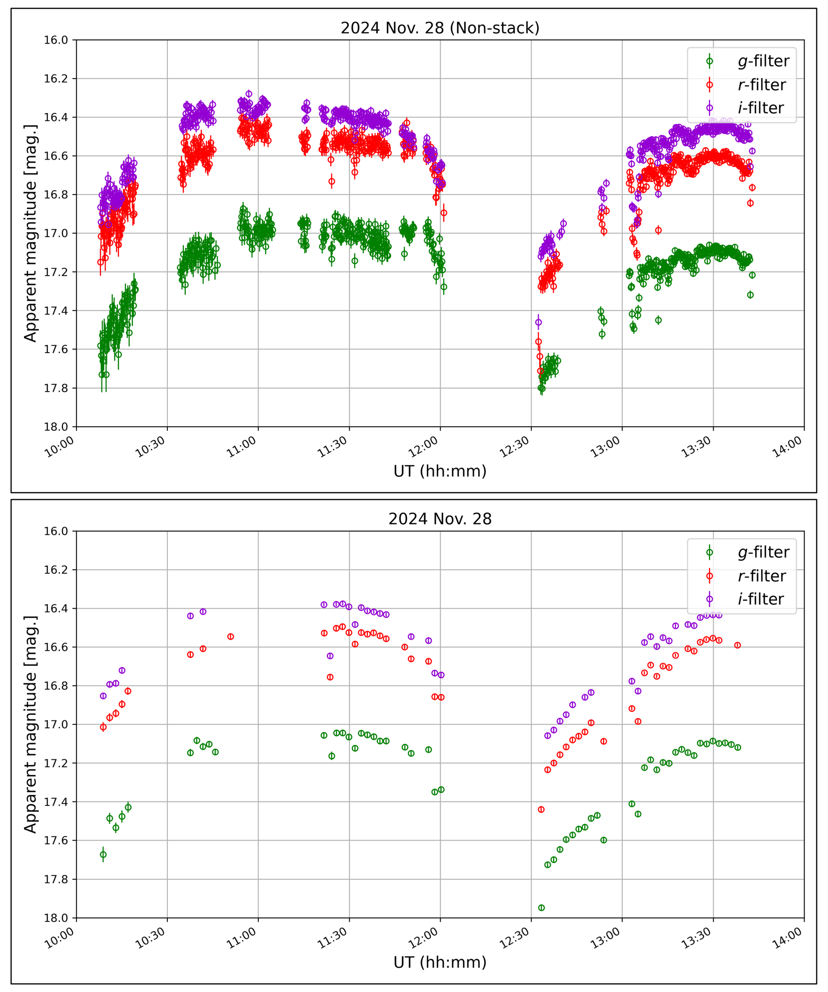


Fig. 8 The multi-color light-curve of Torifune obtained on Nov. 28, 2024. The gaps in the data correspond to periods when observations were not obtained due to bad weather conditions or instrument operation. upper) Non-stacking, achieving a high time resolution with exposure time of 12 s. lower) 10 frames stacking, achieving a high photometric accuracy (S/N > 100).

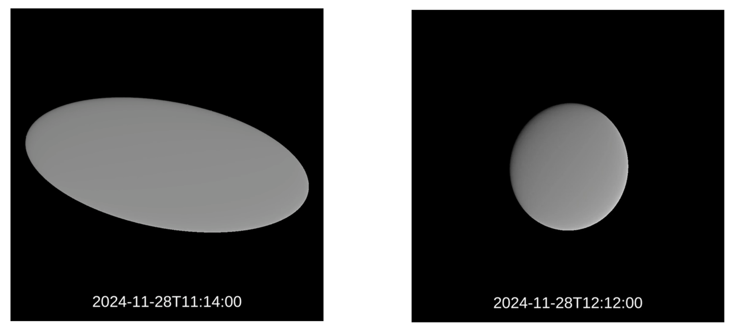


Fig. 9 Geometry model of Torifune on Nov. 28, 2024. The model was constructed using a Hapke model based on the axis ratios and rotation period reported by Popescu+ (2025).

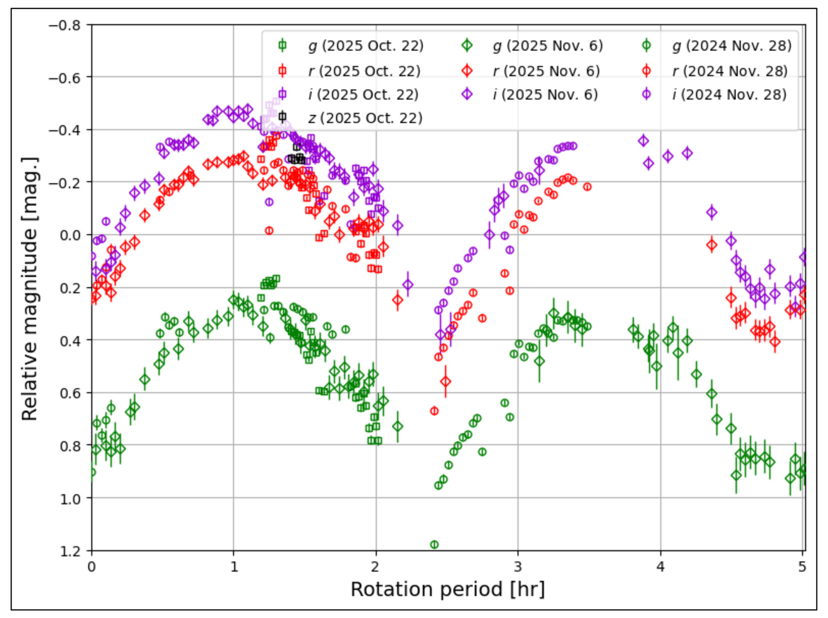


Fig. 10 The multi-color light-curves obtained over the three nights were folded with a rotation period of 5.02 h and overplotted. The magnitudes in each filter are shown as relative magnitudes with respect to the mean value in the *r*-filter.

【Color-index variation】

- Δg-r, Δr-i **~0.05 mag.** < 0.10 mag. (Fig. 11)

→ Small color variation
→ **Globally homogeneous surface color properties**

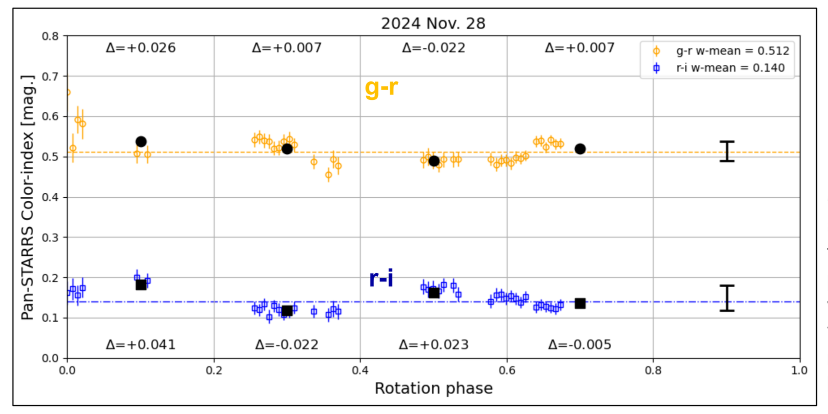


Fig. 11 The rotation period of 5.02 h was converted to a rotation phase ranging from 0 to 1. The rotation phase was further binned with an interval of 0.2, and the variations from the mean values are plotted in black.

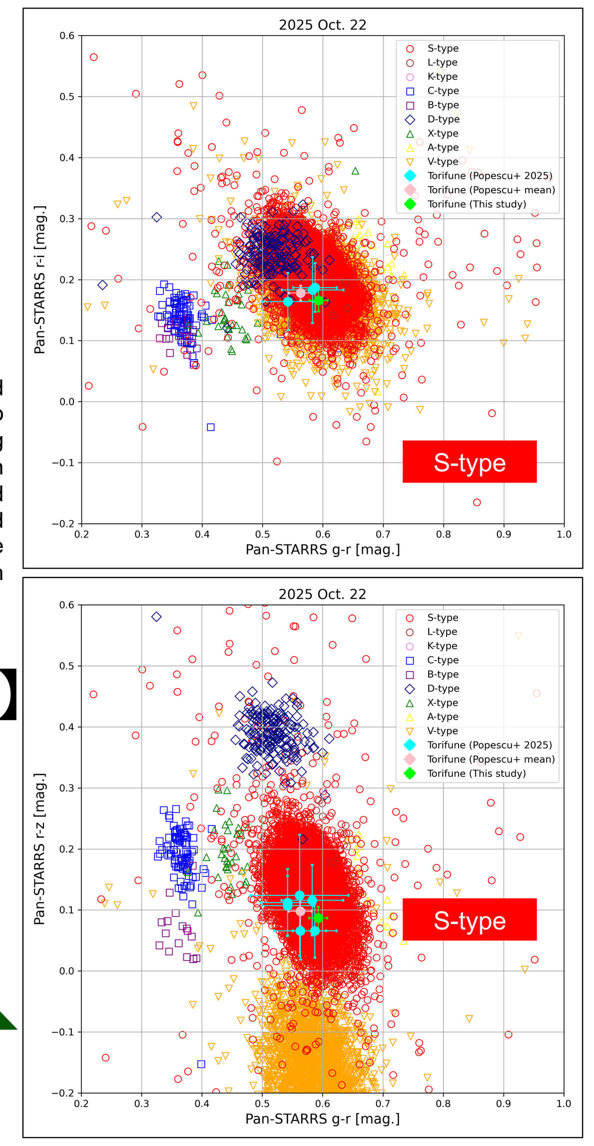
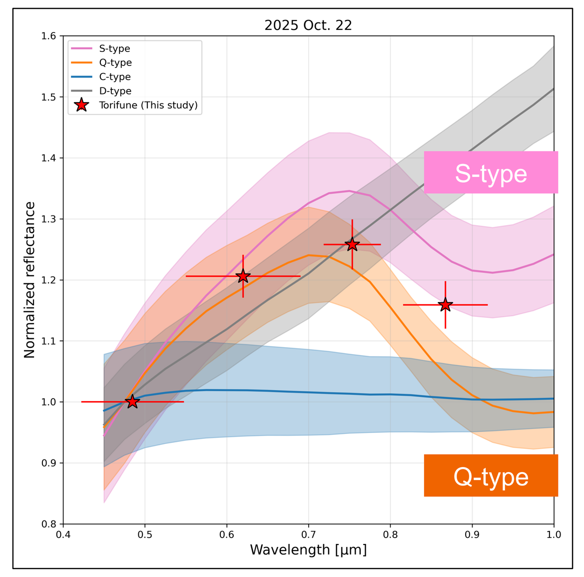


Fig. 12 The mean color-index compared with those of asteroids with known taxonomy types and with the data for Torifune reported by Popescu+ (2025). The data from Popescu+ (2025) were converted from the SDSS photometric system to the Pan-STARRS photometric system using the transformation equations of DeMeo & Carry (2013).

【Mean color-index & Reflectance spectrum】

- **S-type** color-index; g-r, r-i, r-z (Fig. 12)
- Between S-type and Q-type spectrum (Fig. 13)

→ **Sq-type** color properties
→ Moderate degree of space-weathering among S-type asteroids (Fig. 14; Binzel+ 2010)



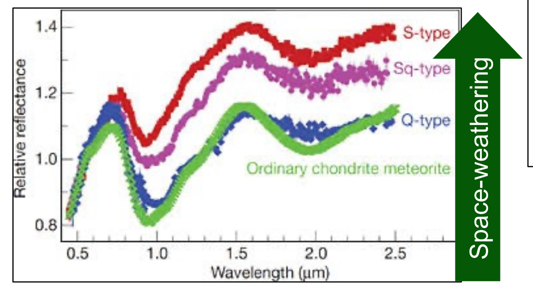


Fig. 14 Visible-NIR S-, Sq-, Q-type reflectance spectrum. Space-weathering is thought to transform the fresh surfaces of Q-type asteroids into the redder (older) surfaces of S-type asteroids (Binzel+ 2010).

Fig. 13 Reflectance spectrum of Torifune derived from the mean color-index (g−r), (r−i), and (r−z) obtained on Oct. 22, 2025. The reflectance values in the *g*-, *r*-, *i*-, and *z*-filters were normalized at the central wavelength of the *g*-filter (0.4849 μm).

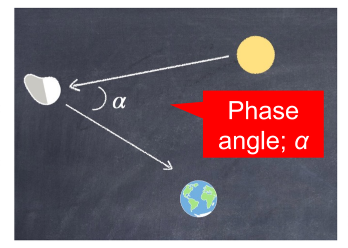


Fig. 15 Phase angle; Sun-Target-Observer.

【Color-index at each phase angle & Phase reddening】

- No significant color variations (< 0.10 mag.) were detected at each phase angle (Fig. 16)
- **S-type** phase reddening trend (Fig. 17)

- Flyby observations at low phase angle; behind the Sun (Hirabayashi+ 2026)

→ May support constrain the onboard camera settings of the spacecraft

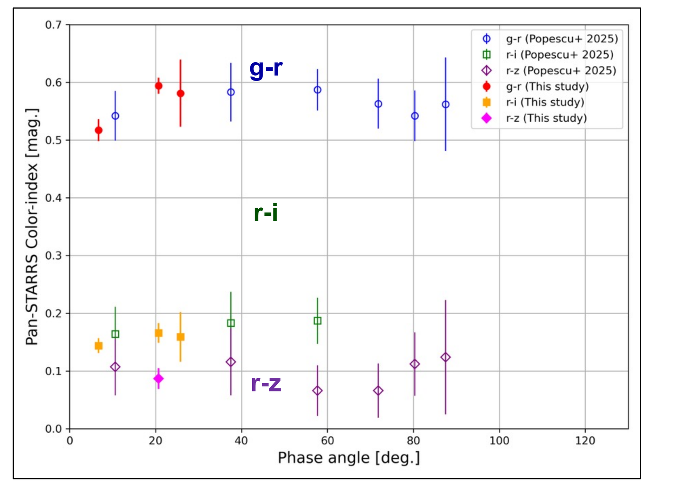


Fig. 16 Mean color-index of Torifune at each phase angle. The SDSS photometric system values reported by Popescu+ (2025) were converted to the Pan-STARRS photometric system. No significant variations larger than 0.10 mag. are observed at any phase angle.

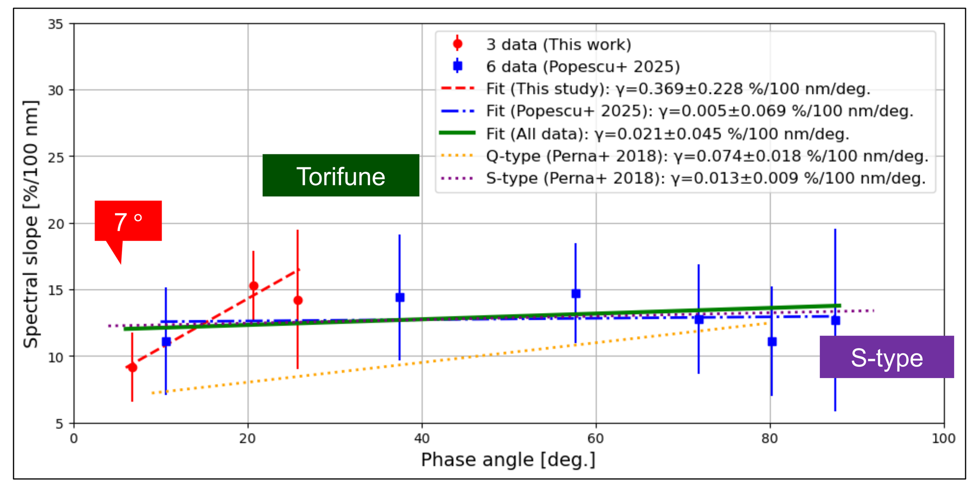


Fig. 17 Spectral slope between the *g*- and *r*-filters as a function of phase angle. The average trends for S-type and Q-type asteroids reported by Perna+ (2018). The green line represents the linear fit to the combined dataset of Popescu+ (2025) and our observations, showing a phase reddening trend consistent with that of S-type asteroids.

4. Summary

- We conducted ground-based observations of the NEA Torifune, a target of the Hayabusa2# flyby mission to investigate its surface color properties
- High time resolution and high photometric accuracy simultaneous three-filter photometric observations were conducted over several nights during 2024-2025

1. The surface color is globally homogeneous
2. The derived color properties are consistent with S-type, particularly Sq-type asteroids
3. No significant color variation is detected at each phase angle. The color data exhibit a phase reddening trend typical of S-type asteroids